

CAN/LIN Frame Generator Plugin for Instrument Studio in LabVIEW

This plugin is an easy-to-use CAN/LIN Bus frame generator. It uses the Measurement Plugin SDK.

Contents

1. The service logic main file is the **CAN-LIN Frame Generator\Measurement Logic.vi**.
2. The UI part is implemented in the **CAN-LIN Frame Generator UI\Measurement UI.vi** file.

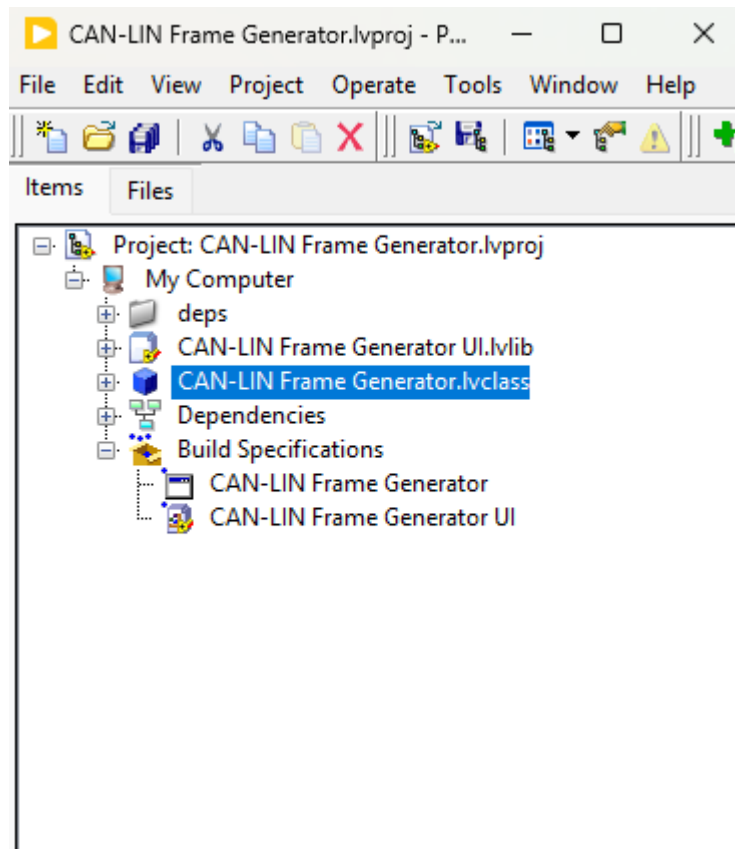
The plugin uses the OOB (Out-Of-Band) gRPC Service for interactive UI part (CONTROL parts of the plugin)

Prerequisites

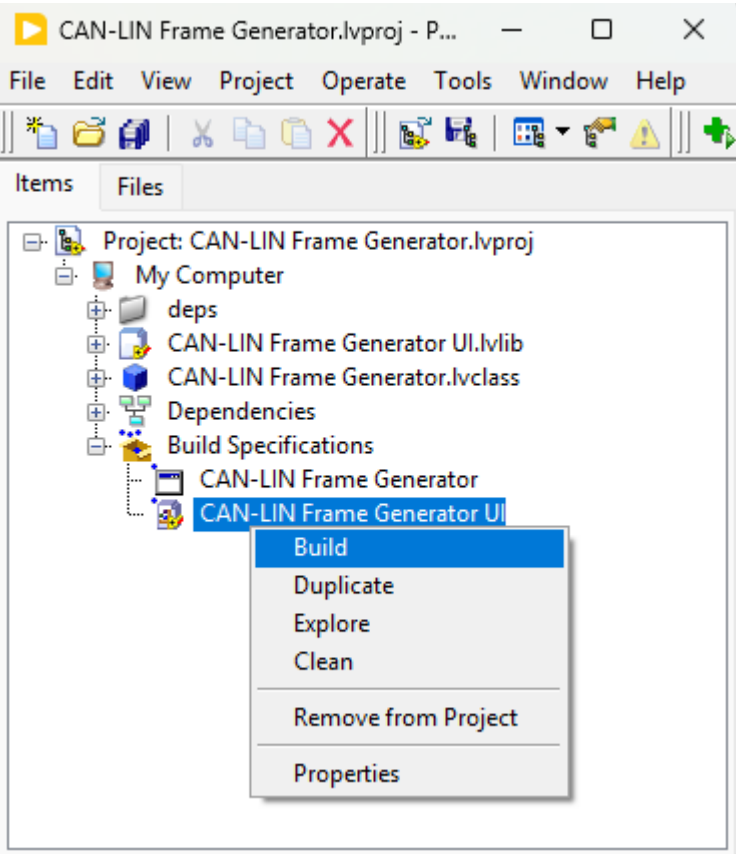
- **LabVIEW 2025 Q3**
- **Measurement Plugin SDK 3.5.0.2**
- **Instrument Studio 2025 Q4.**
- **NI PXIe-8510** with NI-XNET-CAN and/or NI-XNET-LIN transceiver

How to run this plugin

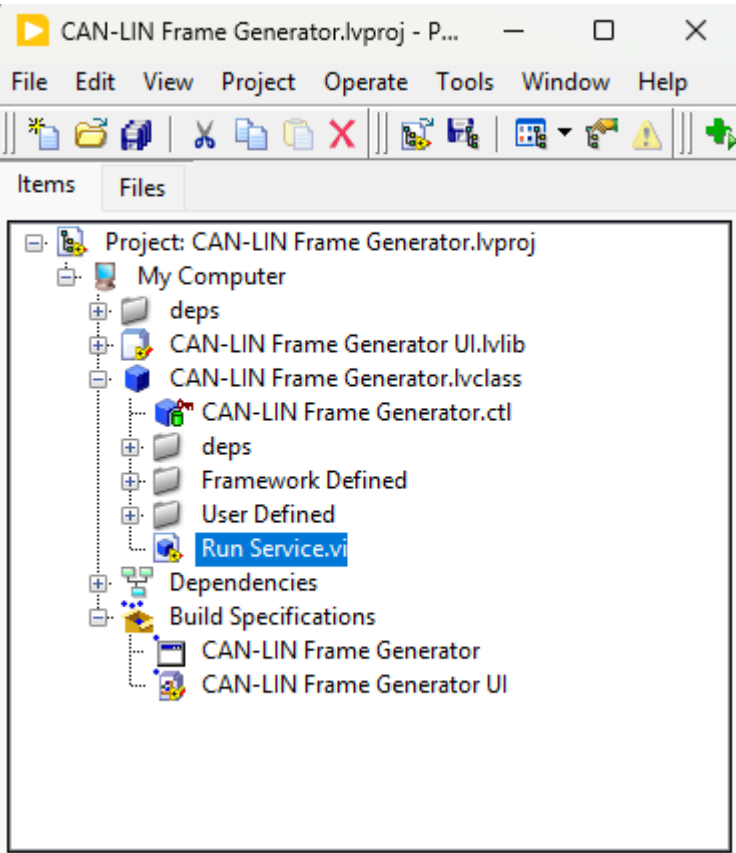
Open the **CAN-LIN Frame Generator.lvproj** project in **LabVIEW 2025**.

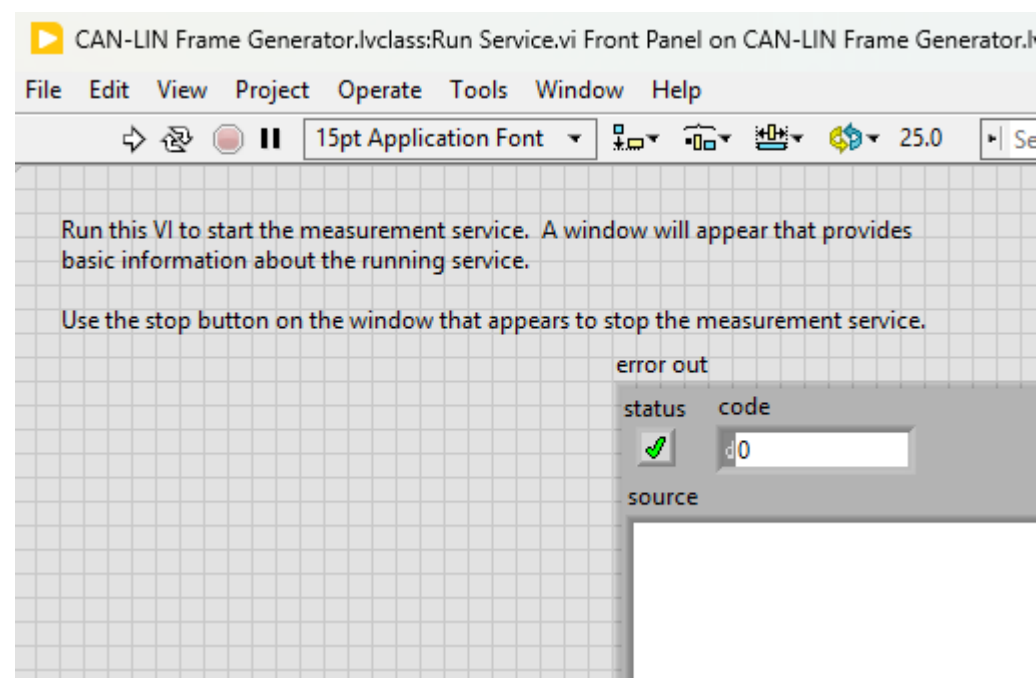


Build the **CAN-LIN Frame Generator UI**.

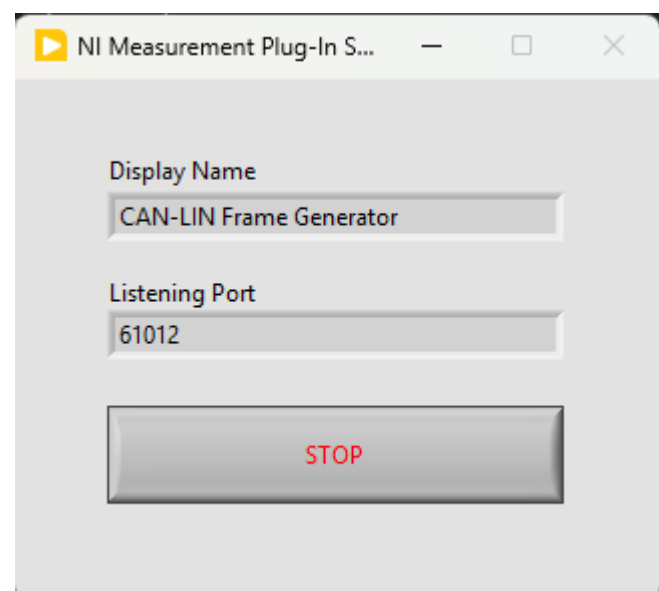


Open **Run Service.vi** from the **CAN-LIN Frame Generator** class

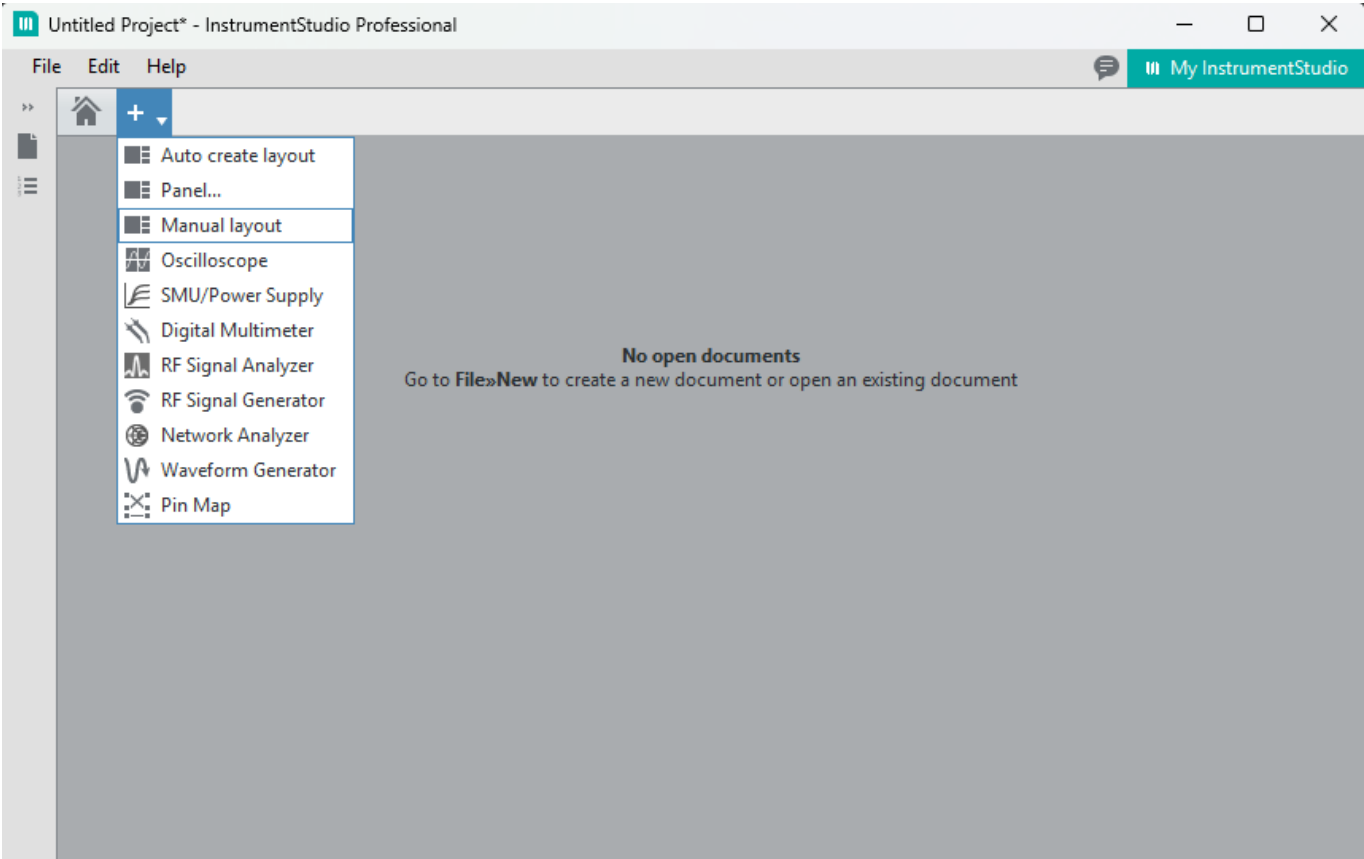




Run the VI

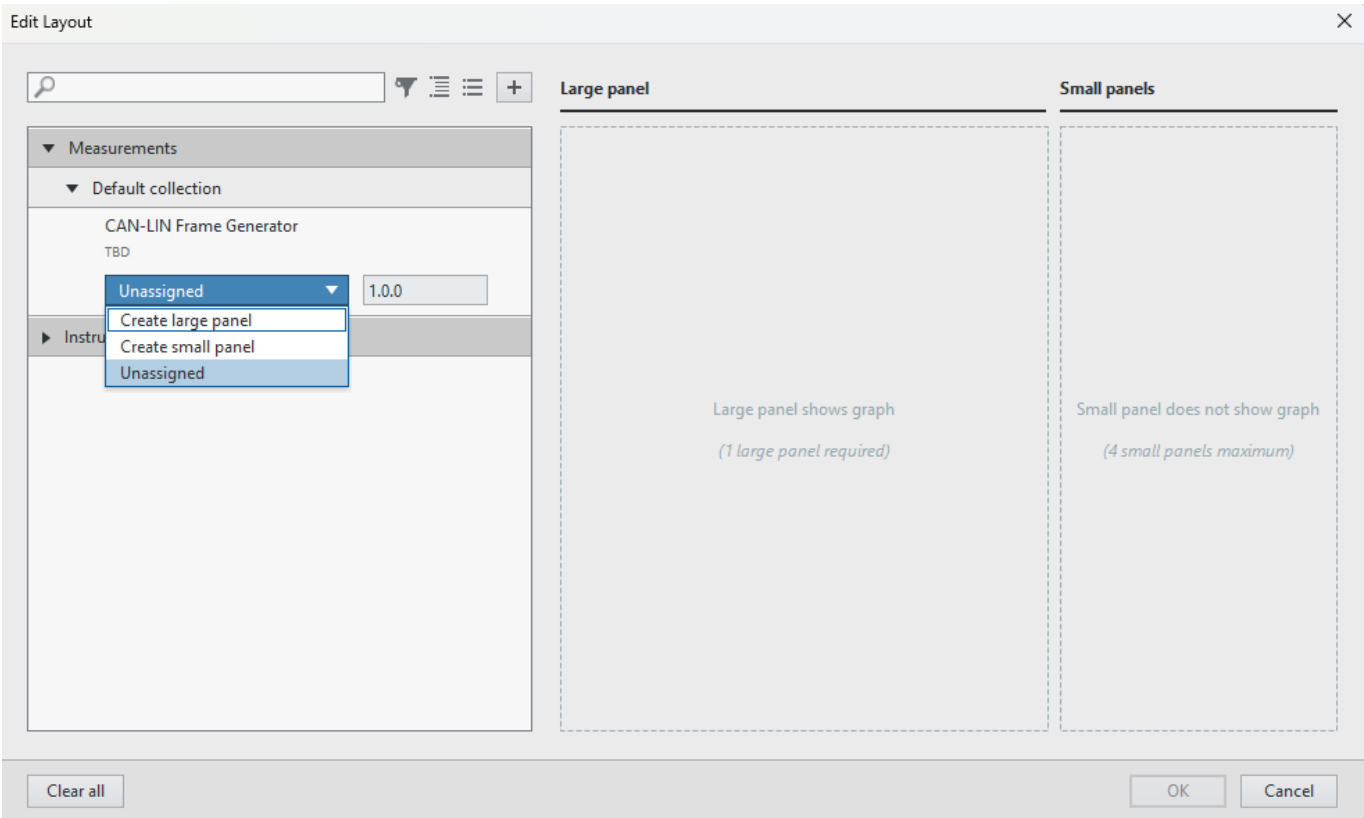


Open **Instrument Studio 2025** and create **Manual Layout**

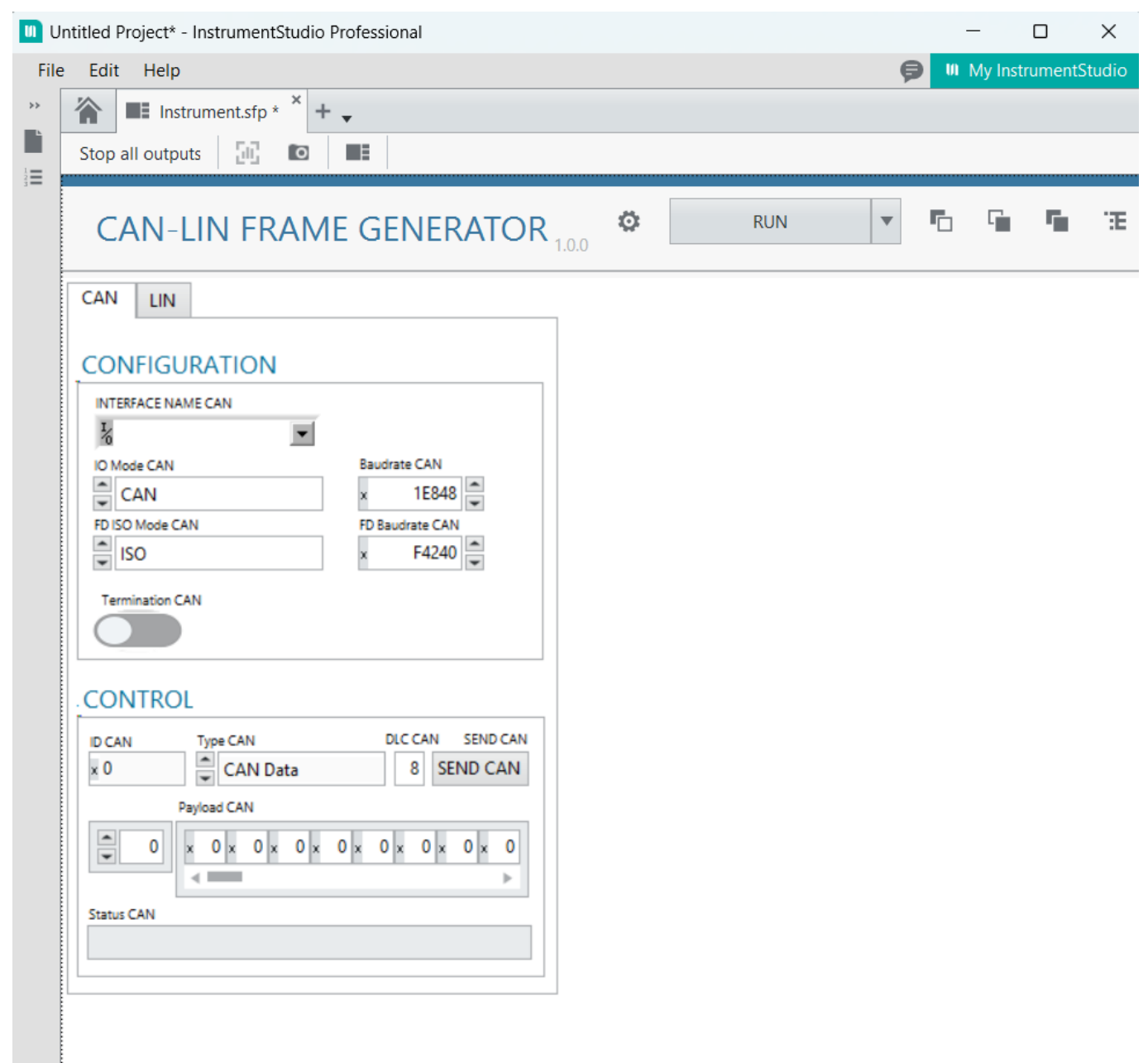


Select **CAN-LIN Frame Generator** as a large Panel

(The plugin can also be run as a Small Panel, but this option is only available alongside another Large Panel, not on its own.)



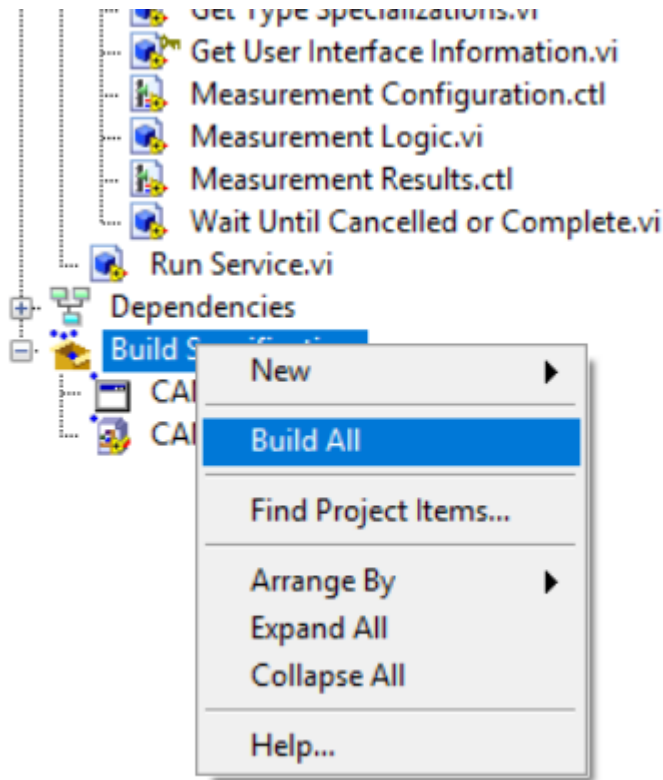
The plugin UI will shown in the Instrument Studio like this:



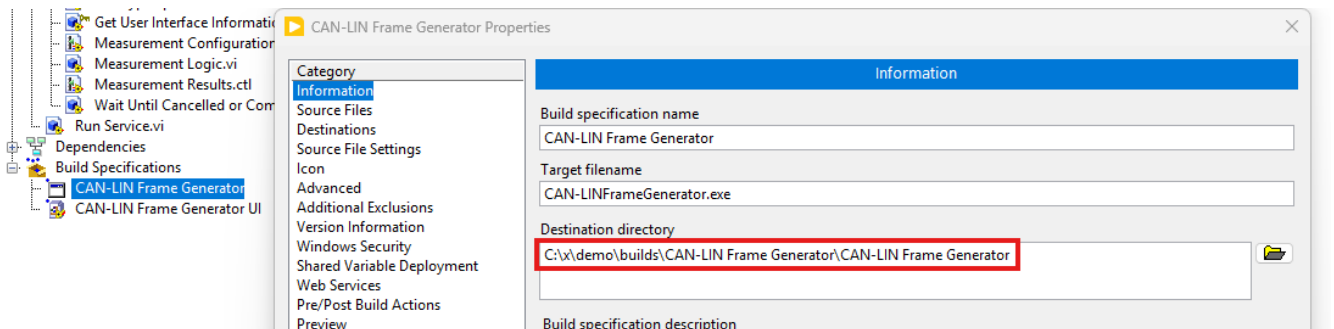
How to deploy the plugin (local only)

You can run the plugin without having to open the project file every single time.

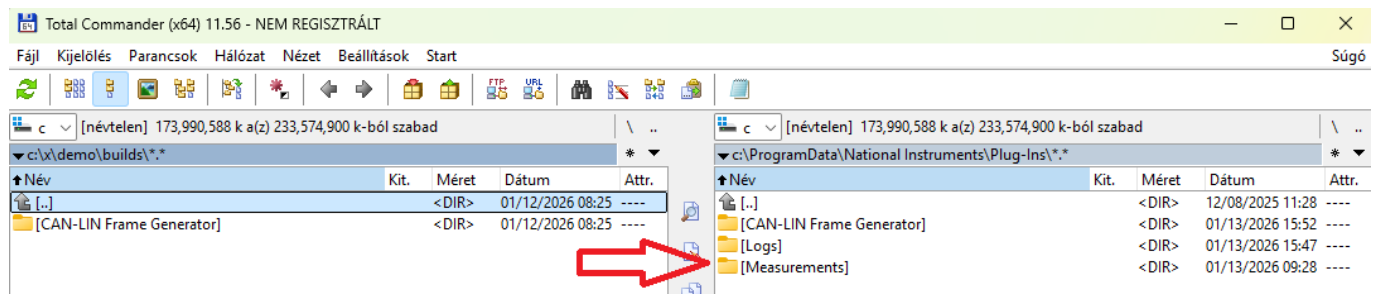
Build the **CAN-LIN Frame Generator** and the **CAN-LIN Frame Generator UI**



Check the name of the Build destination directory.



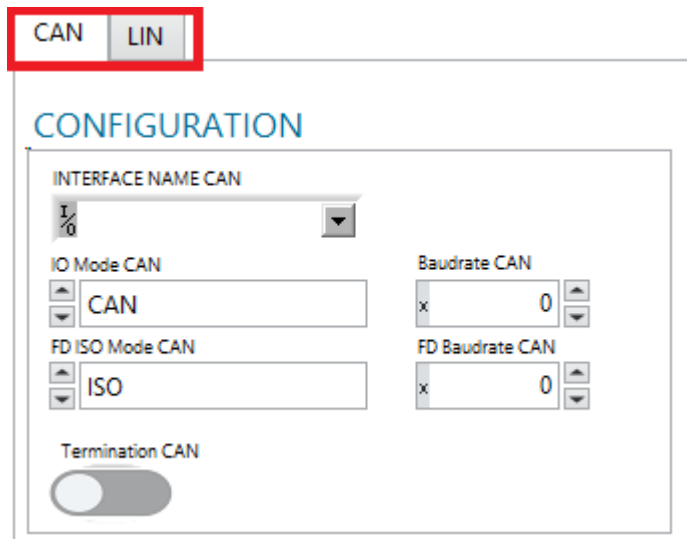
Copy the folder received in the previous step into the `C:\ProgramData\National Instruments\Plug-Ins\Measurements` location (it is a hidden folder).



Now you can run the plugin without the project files being open, just by launching Instrument Studio.

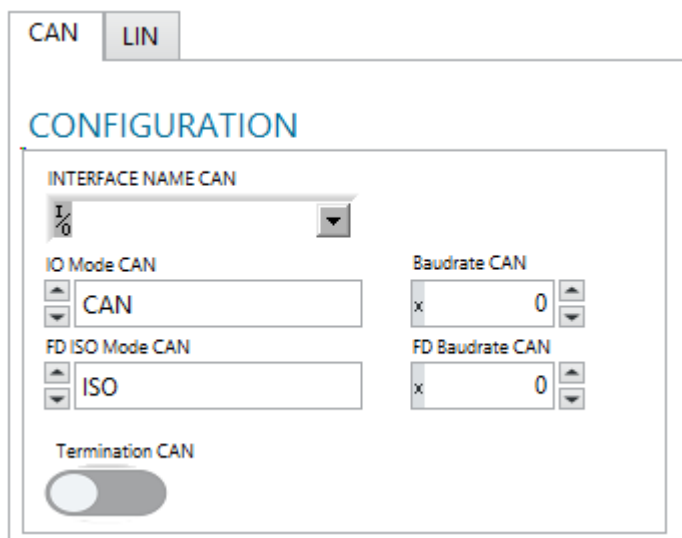
How to configure the plugin

Select the bus type



The screenshot shows a configuration window with two tabs: 'CAN' and 'LIN'. The 'CAN' tab is selected and highlighted with a red box. Below the tabs is a section titled 'CONFIGURATION'. Inside this section, there is a dropdown menu for 'INTERFACE NAME CAN' showing 'I/O'. Below this are two columns of settings. The left column has 'IO Mode CAN' set to 'CAN' and 'FD ISO Mode CAN' set to 'ISO'. The right column has 'Baudrate CAN' and 'FD Baudrate CAN' both set to 'x 0'. At the bottom, there is a 'Termination CAN' toggle switch which is currently turned off.

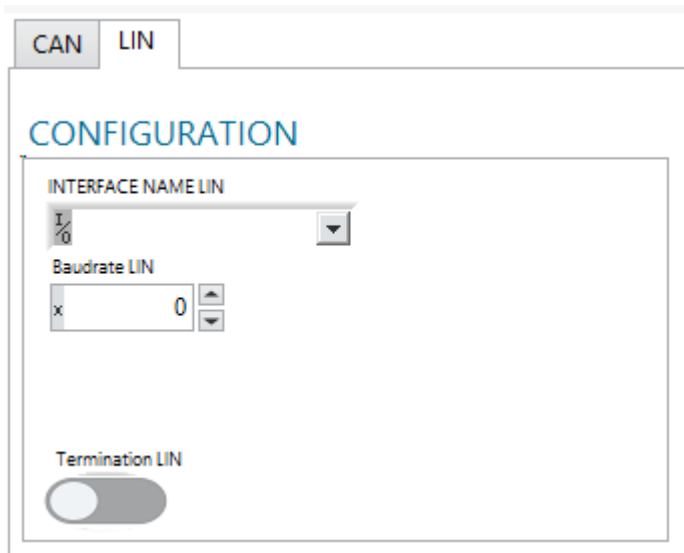
CAN configuration



This screenshot is identical to the one above, showing the same configuration window. However, the 'LIN' tab is now selected, and the 'CAN' tab is no longer highlighted. The configuration settings remain the same: 'INTERFACE NAME CAN' is 'I/O', 'IO Mode CAN' is 'CAN', 'FD ISO Mode CAN' is 'ISO', 'Baudrate CAN' is 'x 0', 'FD Baudrate CAN' is 'x 0', and 'Termination CAN' is off.

1. Select the CAN interface
 - Only valid CAN modules can be selected. The name is the same as the name specified in MAX.
2. Select the IO Mode:
 - CAN (Standard Controller Area Network)
 - CAN FD (Flexible Data-Rate)
 - CAN FD+BRS (Flexible Data-Rate and Bit Rate Switch)
3. Set baudrate (in hexadecimal form)
 - The default value is 1E848 for 125 kbps
4. Set FD ISO Mode (optional when IO Mode is set to CAN FD or CAN FD+BRS)
 - ISO
 - Non ISO
 - Legacy ISO
5. Set FD baudrate (in hexadecimal form) (optional when IO Mode is set to CAN FD or CAN FD+BRS)
 - The default value is F4240 for 1 Mbps
6. Turn "ON" the Termination CAN if it is needed for the bus.

LIN configuration

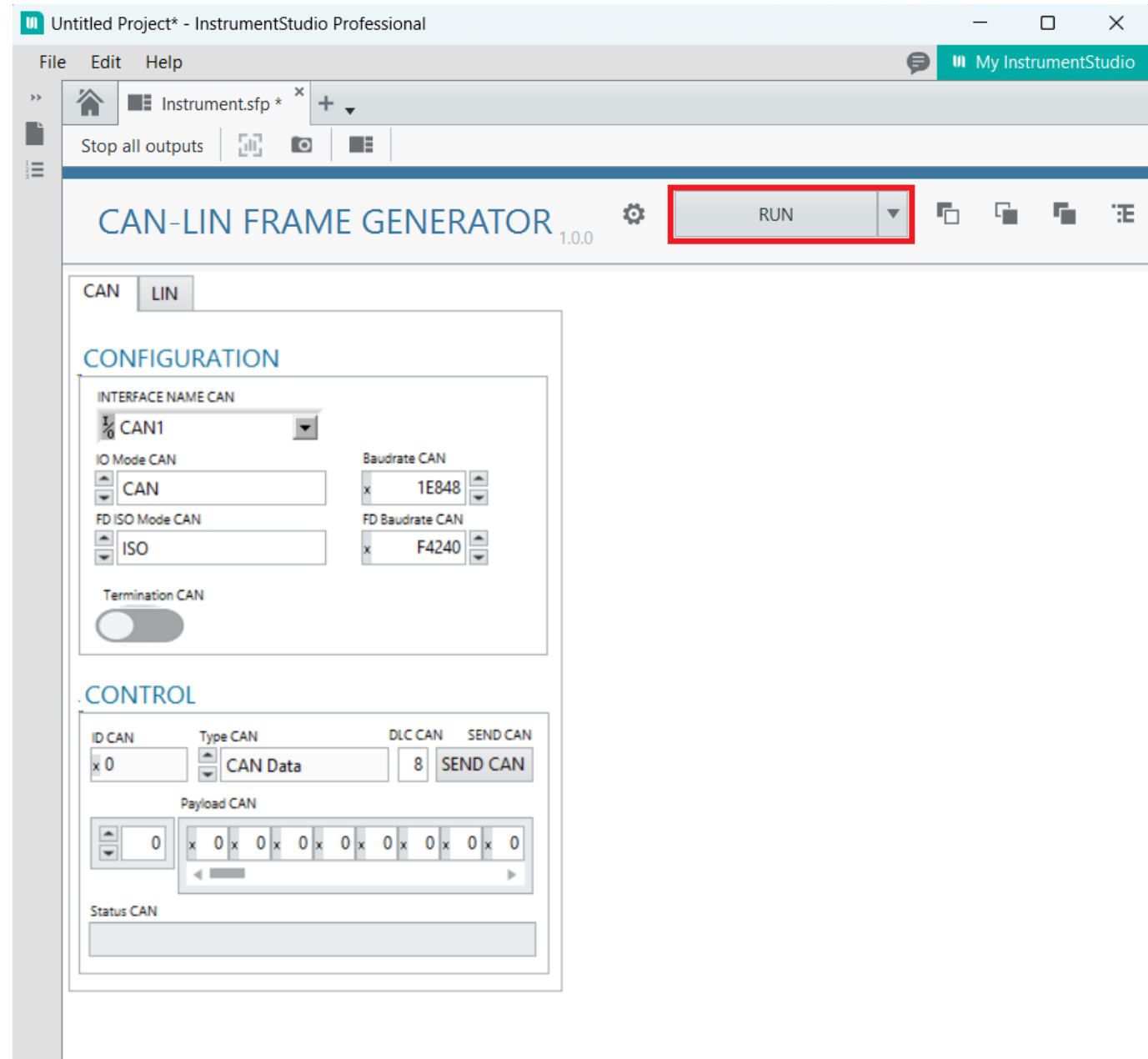


1. Select the LIN interface
 - Only valid LIN modules can be selected. The name is the same as the name specified in MAX.
2. Set baudrate (in hexadecimal form)
 - The default value is 4B00 for 19.2 kbps
3. Turn "ON" the Termination CAN if it is needed for the bus.

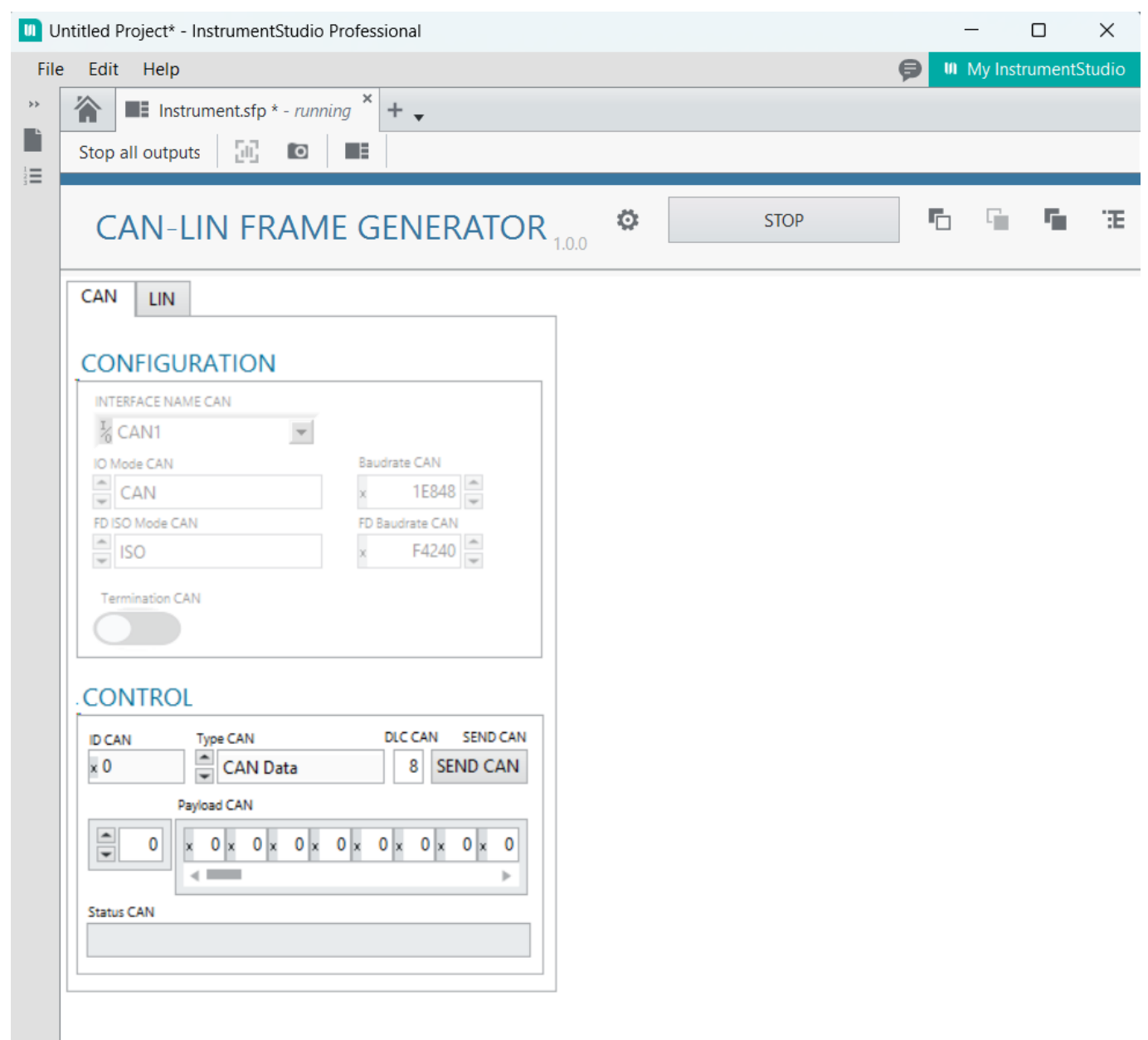
*Note: The plugin connects to the LIN bus as a master node. It can send single Output Stream frames.

How to start and use the plugin

Start the plugin with the **RUN** button in the Instrument Studio.



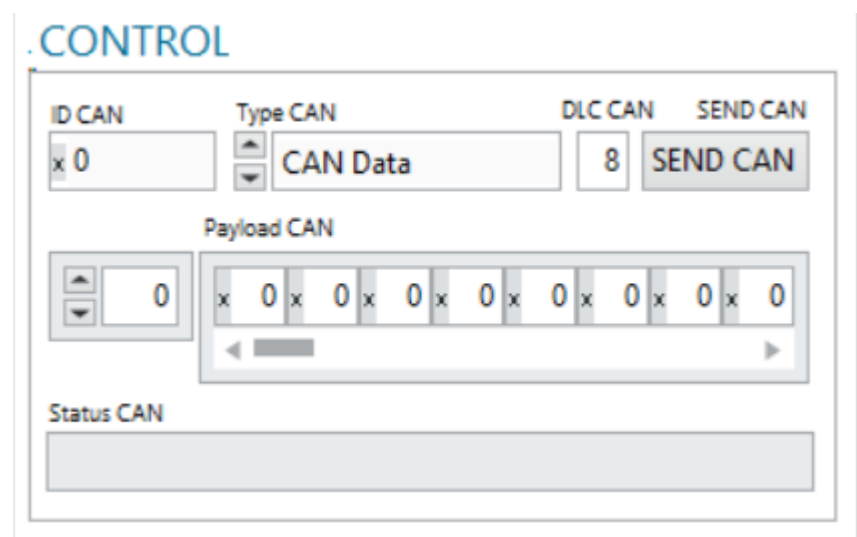
If the configuration was correct, the UI changes as follows:



How to set the output frame and how to send it out

After filling in the CONTROL fields, the user can send the message by pressing the SEND CAN button.

CAN controls



- ID CAN: the identifier of the CAN frame
 - The valid CAN ID range is 0-0x7FF (Standard ID) and 0x800-0x1FFFFFFF (Extended ID). The plugin automatically switches between Standard and Extended IDs
- Type CAN: set the type of frame
- DLC CAN: set the length of the payload in bytes in decimal form
 - In CAN I/O mode the maximum payload length is 8 bytes. For CAN FD, the range can be 0–8, 12, 16, 20, 24, 32, 48, or 64.
 - The default value is 8
- Payload CAN: set the data content of the frame in hexadecimal form
- The Status field indicates the validity of the settings.

You can send out the frame with the SEND CAN button

LIN controls

After filling in the CONTROL fields, the user can send the message by pressing the SEND CAN button.

The screenshot shows a software interface titled ".CONTROL". It contains several input fields and a button:

- ID LIN:** A text box containing the value "0".
- DLC LIN:** A text box containing the value "8".
- SEND LIN:** A button labeled "SEND LIN".
- Payload LIN:** A section containing a scrollable hex input field with eight slots, each containing a "0". To the left of this field is a small numeric input box with the value "0".
- Status LIN:** A large, empty rectangular box at the bottom of the interface.

- ID LIN: set the identifier of the CAN frame in hexadecimal form
 - The valid CAN ID range is 0-0x3F
- DLC LIN: set the length of the payload in bytes in decimal form
 - In LIN mode the maximum payload length is 8 bytes
 - The default value is 8
- Payload LIN: set the data content of the frame in hexadecimal form
- The Status field indicates the validity of the settings.

You can send out the frame with the SEND LIN button